PRODUCT REQUIREMENT DOCUMENT (PRD)

SnapChef — AI Kitchen Inventory & Recipe Synthesizer

1. Product Overview & Vision

Attribute	Specification
Product Name	SnapChef
Platform	Progressive Web App (PWA) / Mobile-Responsive Web Application
AI Engine	Google Argon Model (utilized at maximum potential)
Core Problem	Young adults experience severe decision fatigue when deciding what to cook, resulting in wasted time, food spoilage, or defaulting to food delivery—even when their fridge and cupboards are fully stocked.
Product Vision	A futuristic, high-definition, and highly engaging web application that acts as a smart kitchen inventory manager and global recipe synthesizer. It leverages the Google Argon model at its absolute peak potential to eliminate daily mealtime friction across mobile and desktop browsers.

2. Target Audience

- **Demographic:** Young adults aged 22–30 (Gen Z and younger Millennials).
- **Psychographic:** Tech-savvy individuals who appreciate gamified, highly visual interfaces over plain, clinical utility tools. They value health, fitness, and macronutrient tracking, and rely heavily on mobile phones for daily household tasks, yet struggle with the daily mental load of planning breakfast, lunch, and dinner.

3. Design & Aesthetic Guidelines

The user interface must be high-definition, intuitive, and completely avoid traditional, sterile recipe blog layouts while keeping every navigation option unmistakably clear.

- **Theme:** Modern Cyberpunk / Neo-Tokyo Heads-Up Display (HUD). Futuristic, high-definition visual craft with sharp contrast.
- **Color Palette:** Deep obsidian and matte black backgrounds (#0D0D0D) paired with high-contrast neon accents (electric cyan, acid lime, and magenta).
- **Typography:** Crisp monospace fonts for quantitative metrics (calories, macros, countdown timers) paired with clean, geometric sans-serif typography for effortless step-by-step reading.
- **UI Components:** Glassmorphism panels, glowing borders, subtle terminal hover states, and RPG-style item cards for scanned ingredients.

- **Mobile-First Accessibility:** Built with responsive layouts and touch-friendly controls so the entire site runs smoothly and is immediately accessible on mobile phones.

4. Core Features & Functional Requirements

4.1. Browser-Native Camera Scanner ("The Eye")

- **Immediate Camera Prompt:** Upon initiating the web app via a mobile or desktop browser, the interface immediately prompts for HTML5/WebRTC camera access so the user can scan without navigating through buried menus.
- **Live Viewfinder & Capture:** Displays a live camera viewfinder with a cyberpunk-style targeting reticle. Users point their phone inside their refrigerator, vegetable basket, or kitchen cupboards and capture a photo.
- **Physical Ingredient Mapping:** Sends the captured image directly to the Google Argon model to detect, classify, and catalog all visible perishables, vegetables, proteins, grains, and spices.

4.2. Deep Ingredient Codex & Nutritional Profiler

- **Interactive Inventory Grid:** Scanned items populate a visual "Inventory Loot" grid that users can tap to inspect or adjust.
- **Complete Source Material Breakdown:** Selecting any scanned item (e.g., a basic potato) opens a detailed codex panel displaying:
  - **Source Material Details:** Botanical or culinary origin, classification, and basic composition details.
  - **Nutritional Profile:** Exact macronutrient breakdown (Calories, Protein, Carbohydrates, Fats) per standard 100g serving.
  - **Health Benefits:** Specific micronutrients and physiological benefits (e.g., potassium for muscle recovery, resistant starch for digestion, Vitamin C).

4.3. Global Recipe Synthesizer (Powered by Argon)

- **Meal-Time Intent Filters:** Clear, prominent toggles allow users to filter recipes specifically for **Breakfast**, **Lunch**, or **Dinner**.
- **Zero-Missing-Ingredient Logic:** Cross-references the scanned fridge and cupboard inventory to generate logical, practical, easy-to-cook recipes requiring zero missing core ingredients.
- **Global Cuisine Diversity:** Synthesizes authentic everyday dishes across world cuisines, including:
  - **Indian Cuisine:** Poha, Aloo Jeera, Sabzis, Dals, Parathas, and quick home-style curries.
  - **European Cuisine:** Rustic pastas, Mediterranean bowls, frittatas, and Spanish tortillas.
  - **South American Cuisine:** Arepas, quick bean stews, and fresh ceviche-style preparations.
- **Recipe Summary Metrics:** Each suggested recipe clearly displays preparation and cooking time, total estimated calories, protein content, and a gamified difficulty rating (Level 1 to Level 5 Craft).

4.4. RPG-Style "Crafting" Mode (Interactive Cook View)

- **Step-by-Step Flashcards:** Selecting a recipe launches an interactive cooking HUD that presents instructions as single, focused cards or a vertical timeline rather than dense paragraphs.
- **Integrated Countdown Timers:** Any cooking step involving a specific duration (e.g., boiling, sautéing, simmering) includes a one-tap countdown timer with an audio alert.
- **Session Inventory Update:** Completing a dish updates the user's virtual pantry so remaining ingredients stay accurate for the next meal.

5. User Journey Flow

Stage	Action & System Response
1. Launch	User opens SnapChef on their mobile browser; the app immediately prompts for camera access.
2. Scan	User photographs their open fridge, vegetable drawer, or pantry shelf.
3. Inspect	Argon extracts all ingredients into the Inventory Grid. User taps items to view source details, calories, and health benefits.
4. Filter	User taps **Breakfast**, **Lunch**, or **Dinner**.
5. Synthesize	SnapChef presents logical recipes across Indian, European, South American, and global cuisines using only the scanned items.
6. Craft	User selects a dish, follows step-by-step instructions with built-in timers, and completes the meal.

6. Technical Architecture & Peak Argon Utilization

- **Google Argon Model Integration:** Configured to operate at maximum capability—combining Real2Sim physical spatial recognition to identify overlapping groceries from real-world camera captures, deep multi-cultural culinary reasoning for logical recipe synthesis, and 1M-token context retention to track full kitchen inventories and nutritional profiles in a single session.
- **Frontend & Camera Stack:** Responsive HTML5, CSS, and JavaScript web application utilizing `navigator.mediaDevices.getUserMedia` (with file-capture fallback for broad mobile browser compatibility) and LocalStorage/IndexedDB for persistent pantry state.
- **Structured JSON Output Schema:** The system prompt enforces a strict JSON response payload so the frontend renders ingredient cards, nutritional codex entries, and recipe timers reliably:

```
{
  "inventory": [
    {
      "name": "Potato",
      "category": "Vegetable",
      "calories_per_100g": 77,
      "macros": { "protein_g": 2.0, "carbs_g": 17.5, "fat_g": 0.1 },
      "health_benefits": ["High in potassium", "Source of Vitamin C and B6",
"Contains digestive resistant starch"],
      "source_details": "Starchy tuberous root vegetable native to the Americas,
widely cultivated as a global dietary staple."
    }
  ],
  "recipes": [
    {
      "title": "Aloo Jeera (Cumin Spiced Potatoes)",
      "cuisine": "Indian",
      "meal_type": "Lunch",
      "prep_cook_minutes": 20,
      "total_calories": 250,
      "protein_g": 4.5,
      "crafting_difficulty": "Level 1",
      "steps": [
        { "instruction": "Boil diced potatoes until tender.", "timer_seconds": 720
},
        { "instruction": "Temper cumin seeds in hot oil and toss with potatoes and
spices.", "timer_seconds": 180 }
      ]
    }
  ]
}
```